Lead exposure and IQ loss in children: a Bayesian and frequentist meta-analysis of recent epidemiological studies

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- Journal: Population Health Metrics, BMC
- Using PRISMA 2020 checklist for structure
- Using raRoB tool for Risk of Bias Assessment

Abstract

Background

- ☒ Intro: Lead, health effects, burden of disease
- ☒ Currently used ERFs
- ☒ Rationale for methodological comparison (BMA - FMA)
- ☒ Aim of this study

Lead (Pb) is a pervasive environmental pollutant originating from anthropogenic and natural sources. Although concentrations have declined globally in past decades following the phase-out of leaded gasoline, lead remains present in the environment and human blood (1,2). In Europe, primary exposure sources include dietary products, drinking water, and indoor dust (3,4). The most recent Global Burden of Disease Study estimated that in 2023, 3.4 million (95% Uncertainty Interval (UI): 2.7; 4.0 million) Disability-Adjusted Life Years (DALYs) were attributable to lead exposure in the European Union (EU) (5). This highlights the continued public health significance even of low-level exposure.

Lead exposure is associated with multiple adverse health outcomes across the human life course, including decreased kidney function (6), liver disease (7), coronary artery atherosclerosis (8), cardiovascular mortality (9), infertility in women (10), attention-deficit/hyperactivity disorder in children (11), and antisocial behavior (12). Among these endpoints, impaired cognitive development in children — manifested as reduced IQ scores — (13–16) represents a particularly critical endpoint for public health policy. Children are especially vulnerable to lead's neurotoxic effects due to their developing nervous systems and relatively high exposure per body mass. Because lead-associated IQ decrements are irreversible and persist throughout life, they lead to substantial and enduring societal costs (17), making continued efforts to accurately characterize the exposure-response relationship important for evidence-based policy.

Contemporary blood lead levels (BLL) in the EU have declined markedly and now predominantly fall below 5 µg/dL (HBM4EU Dashboard), well below the concentrations examined in many earlier studies (<10 µg/dL). Importantly, there is no evidence for a threshold concentration below which the effects of lead on the IQ can be assumed to be negligible (18,19). There is evidence for a supralinear exposure-response relationship with steeper IQ decrements per unit increase in BLL at low concentrations compared to higher concentration ranges (16,20). These findings highlight the need for an updated quantitative synthesis that reflects both recent epidemiological evidence and the lower exposure range now predominant in European populations, incorporating recent data to inform risk assessment at currently relevant exposure levels.

Existing burden of disease assessments at national and global scales (5,17,21–23) rely predominantly on two foundational studies: a pooled analysis of seven cohort-studies by Lanphear et al. (2005) or the subsequent re-analysis by Crump et al. (2013) (16,24). While these landmark studies have substantially shaped research and policy on childhood lead exposure, methodological issues and errors in the Lanphear et al. 2005 article have been raised (24,25) and the stepwise backward-elimination regression approach used by Crump et al. (2013), which is central to their findings, has received substantial criticism (26–28). These limitations, combined with the availability of more recent epidemiological data, underscore the need not only for an updated evidence synthesis but also for methodologically robust approaches that can provide reliable exposure-response estimates for burden of disease and risk assessments.

Methodological advances in meta-analysis offer an opportunity to enhance the utility of synthesized estimates for burden of disease and risk assessment applications. While frequentist meta-analysis has been a conventional approach in environmental epidemiology, Bayesian meta-analysis provides a coherent framework for uncertainty quantification through posterior distributions that inherently incorporate both within-study and between-study variability (29). Unlike frequentist confidence intervals, which are often misinterpreted as probabilistic (30), Bayesian credible intervals directly represent the probability distribution of the parameter of interest, facilitating transparent communication of uncertainty. Furthermore, Bayesian posterior distributions can be directly propagated into probabilistic burden of disease and risk assessments, providing seamless integration with probabilistic frameworks in public health decision-making. By comparing frequentist and Bayesian approaches, we will explore how inferential framework and uncertainty quantification influence exposure-response estimates and their application.

The aim of this study is twofold: first, to synthesize and quantitatively pool available epidemiological findings on the association between blood lead levels and IQ scores in children; and second, to conduct both a frequentist and Bayesian meta-analyses, comparing how these analytical frameworks influence exposure-response estimates, uncertainty characterization, and applicability to burden of disease and risk assessment.

Methods

- ☒ Overview: systematic review and meta-analysis, comparing a frequentist to a Bayesian approach
- ☐ Inclusion & exclusion criteria, databases, search strategy, screening process
- ☒ Risk of bias analysis method
- ☒ Description & rationale of harmonization of reported effect measures
- ☐ Description of meta-analysis methodology (frequentist & Bayesian), models and software used, heterogeneity identification
- ☐ Sensitivity analysis

Literature Review

This review adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (31). The checklist can be found in the supplementary files. The reporting of the Bayesian methods and their results was guided by the Bayesian Analysis Reporting Guidelines (BARG) (32).

We conducted a literature review to identify recent studies investigating associations between prenatal or postnatal exposure to Pb and IQ loss in children. Within the larger project that this study was conducted in, there was another study done on the association of (methyl)mercury and lead on IQ loss for different exposure timeframes. Therefore, one joint literature review was conducted for both of these associations. Here, we will only report results for the literature on concurrent lead exposure and IQ loss, but the flowcharts and supplementary material does contain parts of the serach for the other associations and exposure timeframes.
We searched for articles published between January 2005 and July 2023. We only considered studies published after 2005, given the previously-used ERFs in the literature were published before 2005. (source!) Search queries in Pubmed and Scopus included the following terms: substance (Pb - including Mesh terms), outcome (IQ and neurodevelopment), population (children) and results (association, ERF) (Table 1). We prioritized references naming substance (the word "substance"? - to be confirmed) in the title in order to decrease the volume of results. All titles and abstracts were screened by two reviewers. Any conflicts were resolved by a third person. Full text was then assessed by one reviewer. Papers were considered relevant according to predefined PECOTS – Population (children aged 0 to 11 and , pregnant women and women of childbearing age), Exposure (biological measures of Pb, MeHg or Hg from all matrices), Comparator (no exposure or lower levels of exposure), Outcome (IQ or similar indicator), Timing of exposure (no restriction) and Setting (no restriction) (Table 2). We excluded reviews and background articles that inventoried the literature with no new results. We also excluded references written in languages other than English.
From the papers considered relevant after the two-round screening, we collected several descriptive key data from these papers using a grid based on the one proposed by NTP OHAT (2015). When mercury speciation was not clearly stated, we considered total mercury to be methylmercury (Berglund et al. 2005).

Data Extraction

For all studies, adjusted effect estimates were extracted and the set of confounders adjused for recorded.

Risk of Bias Assessment

To assess study quality, we conducted a Risk of Bias assessment using the rapid Risk of Bias (RoB) tool with a specific focus on human observational epidemiological studies (33). Each included publication was assessed for RoB in the domains selection, exposure, outcome, confounding, follow-up, analysis, and selective reporting. In total, 15 items were evaluated for each study., The overall RoB for each individual study was then categorized by one researcher as either low, moderate or high, following two different summary metrics of the RoB tool (for more information, see (33)). Each publication was first assessed by one researcher. Subsequently, consensus with a large language model (LLM) was sought. This was done by uploading the list of all raRoB items along with the respective publication. Then, the LLM was promted to assess the study, using the raRoB items and possible answer categories. If there were any deviations in the judgement of the researcher and the LLM, the respective item was revisited by the researcher and they made the final decision.

Harmonization of Effect Estimates

Studies investigating the association between lead exposure and childhood IQ have made different assumptions regarding the shape of the exposure-response relationship. While some studies assume a linear relationship (34,35), this assumption may only be valid at very low concentration ranges (24). Studies that tested multiple shapes of the exposure-response function have concluded that a log-linear relationship, where BLL are logarithmically transformed, provides a more appropriate model (24,36).

The general form of regression model used for a log-linear association is:

$$y = \alpha + \beta \cdot log_b(X) + \varepsilon \tag{1}$$

Where y represents the IQ score, α is the intercept, β is the regression coefficient (slope), X denotes BLL, b is the logarithmic base, and ε is the error term.

Studies have used different logarithmic bases: the natural logarithm ($\log_e$ where $e \approx 2.71828$), base 2, and base 10. To enable pooling of the effect estimates, we harmonized all regression coefficients and their standard error (SE) to the natural logarithm scale ($\log_e$) using the standardization framework proposed by Rodríguez-Barranco et al. (2017) (37). The transformations are fully documented in suppl. file x
Studies that left BLL on the original (untransformed) linear scale were excluded from the meta-analysis, as transformation from linear to log-linear effect estimates could not be justified based on visual inspection (supplement: plots?).

Meta-Analysis Methodology

In order to quantitatively synthesize the available epidemiological literature on the association of children's blood lead levels and their IQ, we conducted a meta-analysis of the results, comparing a Bayesian to a frequentist approach.
In both approaches, the same studies and effect estimates were included and analogous models were chosen to enable comparison betwen the results. We estimated pooled beta estimates for the absolute loss of IQ points per unit increase in $\log_e$-transformed blood lead concentration (µg/dL).
Within studies, results from the most-adjusted models were chosen to enter the meta-analysis;

unadjusted results were not considered.
R version 4.4.2 (2024-10-31) (38) and Positron version 2026.01.0-147 were used for all computations. All of the computational code for this study is publicly available on GitHub.

Frequentist Meta-Analysis

For the frequentist meta-analysis, we employed a random-effects model using the restricted maximum-likelihood estimation from the R package `metafor` version 4.8-0 (39). The I^2 statistic was used as a quantitative indication of heterogeneity.

Bayesian Meta-Analysis

To synthesize results across studies, we employed a Bayesian random-effects meta-analysis model with two hierarchical levels. At the first level, the observed effect size from each study $\hat{\theta}_k$ is modeled as an estimate of that study's true underlying effect θ_k, with deviation from the true effect determined by the within-study sampling error σ_k (treated as known based on reported study statistics). At the second level, these study-specific true effects are themselves assumed to vary across studies following a normal distribution, characterized by an overall pooled effect μ and between-study heterogeneity τ.

$$\begin{aligned} \hat{\theta}_k &\sim \mathcal{N}(\theta_k, \sigma_k) \\ \theta_k &\sim \mathcal{N}(\mu, \tau) \end{aligned} \tag{2}$$

Where:
$\hat{\theta}_k$ = Observed effect size for study k
σ_k = Within-study sampling standard deviation for study k
θ_k = True underlying effect for study k
μ = Overall pooled effect
τ = Between-study standard deviation (heterogeneity)

Priors

- Parameters to be estimated: mu and tau (mu later called beta?)
- choice based on previous evidence, e.g. (14)
- explain scale (log, µg/dL)

$$\begin{aligned} \mu &\sim \mathcal{N}(-1, 2) \\ \tau &\sim \mathcal{N}^+(0, 2) \end{aligned}$$

Sensitivity Analysis

We conducted two sets of sensitivity analyses. To examine the sensitivity of the results to possibly biased study results, we ran both the frequentist and Bayesian models again, excluding (1) studies with a high RoB and (2) with a high or medium RoB.

To examine the sensitivity of the results to the choice of prior distributions, we fit the Bayesian model again twice, with alternative priors for both the main effect (μ) and heterogeneity (τ). The first set of priors was more informative (more narrow) than the main priors, with $\mu \sim \mathcal{N}(-1, 1)$ and $\tau \sim \mathcal{N}^{+}(0, 1)$. The second set of priors was less informative (or broader) with $\mu \sim \mathcal{N}(0, 6)$ and $\tau \sim \mathcal{N}^{+}(0, 6)$. Notice that for the main effect, we centered the broader prior around 0, giving equal weight to the probability of a protective and detrimental effect of lead.

More informative priors (more narrow):

$$\begin{aligned} \mu &\sim \mathcal{N}(-1, 1) \\ \tau &\sim \mathcal{N}^{+}(0, 1) \end{aligned} \tag{3}$$

Less informative priors (more broad):

$$\begin{aligned} \mu &\sim \mathcal{N}(0, 6) \\ \tau &\sim \mathcal{N}^{+}(0, 6) \end{aligned} \tag{4}$$

Results

- ☐ Cite studies that might appear to meet inclusion criteria, but were excluded & explain why
- ☐ Study characteristics (table), including original and harmonized effect measures
- ☐ Flow diagram, description of (full text) screening results
- ☐ RoB results for each included study
- ☐ Forest plots
- ☐ Sensitivity analysis results

Discussion

- ☐ General interpretation of results in light of evidence in literature
- ☐ Discuss differences, similarities, strengths and weaknesses of freq vs. Bayesian approach
- ☐ Study limitations & strengths

Conclusions

- ☐ Conclusion, with reference to aim of study
- ☐ Implications of results: policy, (scientific?) practice, future research needs

List of Abbreviations

Abbreviation	Meaning
BLL	Blood lead levels
CI	Confidence interval
EBD	Environmental Burden of Disease
EU	European Union
IQ	Intelligence quotient
LLM	Large language model
RoB	Risk of Bias
SE	Standard error
UI	Uncertainty Interval
DALY	Disability-adjusted Life Year

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